

Toward a Model of Academic Development: Schooling and the Acquisition of Knowledge

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At times, it is hard to grasp the tremendous investment that American society has made in articulating and measuring the indicators, benchmarks, or standards of knowledge acquisition in the last 50 years. This investment has not simply been fiscal, although the money involved in certifying educational attainment is truly staggering. Without doubt, the emotional, educational, social, and political cost of this enterprise has been equally overwhelming. At the heart of this endeavor is a search for evidence as to what it means to be educated, a desire to document tangible, positive outcomes from the orchestrated experience we call schooling. However, this very enterprise has become a wedge between educators and the public at large. It has also created tensions within the ranks of the educational community. The political and public fascination with such academic markers is difficult for many educators to comprehend or to justify (Feinberg, 1988). These individuals perceive such efforts as detrimental to the very goals of education. But, others see this public outcry for documentation or substantiation of learning as a critical opportunity for reforming and reframing the educational experience (Hirsch, 1996). Wherever one stands on this divisive issue, there is little solace in the current state of education, little sense that schooling is all that it can be for all of America's citizenry.

Perhaps we should not be surprised that those in political and public arenas talk almost obsessively about standards, or turn to measures of accountability or competence, which seem to many in the research community to be

unsoundly cast, as a way to document the effectiveness of learners, teachers, and schools (House, 1996; Wolf, 1998). What benchmarks of "sound" academic development have we, as educational researchers, offered as alternatives? What more coherent and comprehensive picture of learner growth and development has been placed before the public as a reasonable counter to standards or measures considered imperfect or even counterproductive to the educative experience?

If readers are expecting to find a delineation of feasible alternatives to these standards or measures in the pages of this commentary, they will surely be disappointed. Any such delineation would be premature, in my judgment, since the educational community does not have the comprehensive models and theories of academic development needed to frame sound alternatives. That is to say, data regarding powerful instructional approaches or notions of best practice clearly exist in the educational literature (e.g., Bereiter & Scardamalia, 1989; Cognition and Technology Group at Vanderbilt, 1996; Collins, Brown, & Newman, 1989). Yet, these data offer only a glimpse of the long-term changes that students undergo when engaged in formal learning. In fact, educators still have not embraced a concept of academic development. Consequently, as a research community, we have not immersed ourselves in the kind of longitudinal, multidimensional, multimethod studies that would allow us to articulate credible models and theories of academic development.

If you doubt this, look at the myriad of volumes related to teaching and learning. Do you find any discussion of

the systematic changes that students should undergo cognitively, motivationally, and socioculturally as they are becoming formally educated—not just for a year or two but across the life-span? You will likely encounter individual chapters devoted to cognitive, linguistic, motivational, social, and emotional development. Perhaps the idea is that if we understand these separate dimensions of human learning and development, we can make sense of the systematic changes that students experience as a consequence of formal schooling (i.e., academic development). The fallacy in such an assumption should be readily apparent.

Thus, my intention here is to proffer the study of academic development as a viable agenda for the research community, one that can offer a worthwhile alternative to the more disjointed assessment of knowledge acquisition characteristic of the American educational system. However, because many in the public and political sectors continue to equate schooling with the accumulation of knowledge, and because knowledge will likely remain a key component in models and theories of academic development, I want to focus briefly on the theme of schooling and knowledge.

An Epistemological Orientation

Certainly, questions about the association between schooling and knowledge acquisition are not new to educational researchers. In 1979, when I began my doctoral studies, I was assigned a text that altered my thinking about learning and instruction in significant ways and that set me on a course of educational research. That book was *Schooling and the Acquisition of Knowledge* by

Anderson, Spiro, and Montague (1977). The foundation for this work was an interdisciplinary gathering of psychologists, educators, and philosophers held on Shelter Island in San Diego in 1975. These noted scholars came together to ponder the complex relations between schooling and knowledge and to contemplate such perplexing questions as "How is knowledge acquired?"

In the groundbreaking volume that arose from that conference, Richard Anderson, John Bransford, Ann Brown, Allan Collins, David Rumelhart and others sought to bring the concept of knowledge, as conceived through the eyes of cognitive psychologists, particularly information-processing theorists, into conjunction with schooling in the 1970s. Having just left the classroom to seek my fortune in the academy, I was captivated by these scholars' reflections on knowledge and the acquisition of knowledge. In many ways, the theories, themes, and techniques captured in *Schooling* are symbolic of an era of knowledge research that remains unparalleled—an era in which researchers aggressively pursued questions about the structure, acquisition, retention, and influence of knowledge.

It is hard to imagine that more than 20 years have come and gone since *Schooling and the Acquisition of Knowledge* was published. So much has changed since ideas like schema theory entered our vernacular and dotted our educational landscape, even as other conditions and situations have continued in almost an educational time warp (Alexander & Knight, 1993; Alexander, Murphy, & Woods, 1996). Since those days of information-processing theory, the educational community has been introduced to a plethora of new theories about learning and development that place knowledge in far more varied light. We have become acquainted with such theoretical notions as constructivism, social constructivism, radical constructivism, deconstruction, postmodernism, critical theory, social learning, situated learning, cognitive apprenticeships, and situated cognition—to name but a few.

As a result of this montage of new theories and their alternative perceptions of human learning and development, knowledge has again become a focal point of debate. Aspects of this theoretical discourse center on the degree to which knowledge is socially de-

rived or individually formed, and on whether knowledge is internally held or externally triggered. I do not intend to enter into this debate here. I only wish to point out that, whatever one's epistemological leanings, there is now a deeper appreciation of the impact that society, culture, and context continually have on knowing and knowledge (e.g., Greeno and the Middle School Mathematics Through Applications Project Group, 1998; Lave, 1988; Rogoff, 1990; Saxe, 1991) than in years past. Many have come to heed the messages of those who speak of socially shared learning or the distributed nature of knowledge (Resnick, Levine, & Teasley, 1991; Salomon, 1993)—something that would have seemed quite alien decades before. As a consequence of these debates, any ideas we may have had about the simplicity, unidimensionality, certainty, and constancy of knowledge in ages past—all those epistemological conceptions that Perry (1970, 1981) and others (Schommer, 1990, 1993) have documented in the young or unenlightened—have been supplanted with an understanding of knowledge as complex, multidimensional, fallible, and dynamic.

Even in the face of these new understandings and growing sophistication about knowing and knowledge within the educational community, the past research on schooling and knowledge has produced some powerful and replicated findings that should not be ignored. For one, above and beyond the shared understandings that arise from the sociocultural nature of human learning and development, there remains an idiosyncratic quality to knowing and knowledge. Also, these personalized understandings can transcend any specific moment, place, or social milieu (Cobb, 1994). Indeed, even the perception of knowledge offered here was not born out of any single social exchange or any one context. Rather, it stands as a legacy to the thousands of relevant encounters and experiences that have been woven into the tapestry of memory.

Specifically, knowledge as portrayed here can be thought of as a conceptual scaffold, a foundation for subsequent learning that colors and filters one's school and nonschool experiences (Alexander & Murphy, 1998; Alexander, Schallert, & Hare, 1991). So broadly defined, knowledge signifies all that one knows or believes, without direct con-

sideration of the source of that knowledge, its explicitness, or its veracity. Thus, knowledge is not meant to convey some canon of simple factual tidbits or litany of academic "truths" (Hirsch, 1996). Rather, knowledge, from this epistemological vantage point, constitutes the realm of human understandings, whether accurate or incomplete, declarative or procedural, tacit or explicit. This is not to say that this epistemological stance represents consensus among educational researchers, or that questions regarding the source or authenticity of knowledge are unimportant. To the contrary, multiple perspectives on knowledge have always existed in the educational research community and in the public at large, and will continue to do so (Reynolds, Sinatra, & Jetton, 1996). These epistemological differences are nontrivial and help to explain diverse programs of and approaches to research and practice.

Yet, the changes in perceptions of knowledge and the concomitant decrease in attention to internal processing and individual understandings that have occurred since the 1970s cannot justify setting aside the decades of strong and replicable evidence that one's knowledge is a powerful and pervasive force in learning. As evidence of its power and pervasiveness, knowledge has been repeatedly and positively aligned with such critical processes and outcomes as the following:

- individuals' perspectives and perceptions (e.g., Alexander & Dochy, 1995; Pichert & Anderson, 1977);
- the allocation of attention (e.g., Reynolds, 1992; Reynolds & Shirey, 1988);
- subsequent academic achievement (e.g., Ackerman & Heggstad, 1997; Stanovich, 1986);
- comprehension and memory (e.g., Alvermann, Smith, & Readence, 1985; Anderson, Reynolds, Schallert, & Goetz, 1977);
- learners' motivations, including their interests, goals, and self-efficacy (e.g., Bandura, 1993; Hidi, 1990; Meece & Holt, 1993; Pintrich & Schunk, 1996);
- strategic processing and problem solving (e.g., Alexander & Judy, 1988; Harris & Graham, 1996; Reimann & Schult, 1996; Weinstein & Mayer, 1986);

- openness to persuasive arguments or convincing evidence (Murphy, 1999; Petty & Cacioppo, 1986); and,
- judgments of relevance or importance (Jetton & Alexander, 1997; Stahl, Hare, Sinatra, & Gregory, 1991).

Not only is knowledge, in association with other processes and outcomes, of consequence, but the very quest for knowledge or understanding is also a valued goal, as well as a personally motivating and aesthetically enriching undertaking (Alexander, 1997; Csikszentmihalyi, 1990). Many reading these words have dedicated much of their lives to the pursuit of knowledge and to the sharing of such knowledge in teaching and in writing. Thus, their respect for knowledge has a long and rich history—a heritage that is to be honored.

While not failing to pay tribute to the everyday, informal, and incidental encounters that have unquestionably shaped our perspectives and contributed dearly to our knowledge base (Alvermann, 1998), subsequent discussion focuses on the realm of formal or intentional learning (Bereiter & Scardamalia, 1989). Likewise, such formal learning is considered in relationship to the socially constructed and politically sanctioned organizations called schools. I want to revisit the mission of that San Diego conference of 1975—contemplating the relationship between schooling and knowledge. In so doing, I pose the following question:

Has the last quarter century of theoretical and philosophical deliberations and educational inquiry endowed us with a new-found wisdom as to the nature of formal learning and the systematic, long-term effects such learning has on those who participate in that experience?

In some ways, this question echoes the words of Robert Gagné in his epilogue to that historic conference:

If we set aside, for the moment, our concerns with the questions of how people learn and look simply at the *outcomes of schooling*, what do we see? Are students acquiring enough knowledge, or the right kind of “knowledge”? Or, in a more nearly ultimate sense, are they becoming more competent, more able to adjust to and cope with living in our complex society, more capable of practi-

cal problem solving, increasingly able to find a satisfying quality to their lives? (1977, p. 411).

My personal reaction to Gagné’s intriguing questions is that we still know regrettably too little about the course of formal learning across the lifespan and about the anticipated changes that occur in those who set out on that course. Therefore, let me throw down the gauntlet to the research community and call for greater systematic, multidisciplinary, and multidimensional study of the formal educational experience and the changes, cognitive and noncognitive, that this experience evokes in its participants—what I refer to as academic development. It would be my sincere hope that this systematic, multidisciplinary, and multidimensional research would result in new models of academic development that are both coherent and comprehensive.

Academic Development

What is Academic Development?

Let me elaborate further on this concept of academic development. Academic development, as with other forms of development, refers to systematic change over time. In the case of academic development, the focus of that transformation is centered on the process of formal learning and on the outcomes that have been termed “schooled knowledge” or “scientific concepts” (Alexander, 1992; Gardner, 1991; Vygotsky, 1934/1986). Of course, formal learning does not take place only in the classroom (Cole, 1996). Yet, schools are institutions with an express educational mission (Dewey, 1916/1944), and one in which long-term participation is mandated. For models of academic development to be coherent and comprehensive, they should possess certain characteristics, reflective of current wisdom on learning, development, and instruction. First, they should represent changes that transpire in individuals over the lifespan of their formal learning (Collins et al., 1989). Consequently, even though we may, as a society, concentrate resources on the schooling of our children and youth, formal learning for many—certainly for members of the educational research community—continues throughout one’s lifetime. Models of academic development must, therefore, accommodate such continued growth and

allow for an interface between formal learning prior to, during, and beyond the preK–12 school years.

In a related sense, while contrasts between novices and experts in particular domains have been informative as to certain cognitive consequences of formal learning (e.g., Chi, Glaser, & Farr, 1988; Ericsson & Smith, 1991), such contrasts do not capture the gradual and less dramatic transformations that are likewise central to academic development. Consequently, models of academic development must embrace all learners who undergo the educative process, including those who are neither truly novice nor clearly expert—in effect, the vast majority of individuals participating in this process (Alexander, 1997). Moreover, from a methodological standpoint, a broad and deep understanding of academic development cannot be achieved *solely* through the cross-sectional investigations that have been the mainstay of experimental research (Kulikowich, Brown, & Bokoros, 1998). Rather, systemic, long-term, longitudinal investigations must be undertaken at various points in the educative process, including childhood, adolescence, and adulthood.

Second, since the publication of *Schooling and the Acquisition of Knowledge*, there has been a heightened awareness that learning is not a “coldly” cognitive phenomenon, but encompasses motivational dimensions and is reflective of the social/cultural environment (Brown, Bransford, Ferrara, & Campione, 1983; Pintrich & Schrauben, 1992). This is precisely why the study of academic development cannot be restricted to the study of cognition in isolation from other developmental forces. In the work on domain learning in which my colleagues and I are engaged, we explore formal learning not only in terms of the relevant knowledge that individuals of varying levels of competence convey, but also in terms of the quantity and quality of their strategic processing, as well as the interests they manifest (e.g., Alexander, Jetton, & Kulikowich, 1995; Alexander & Murphy, 1999; Alexander, Murphy, Woods, Duhon, & Parker, 1997). Although I personally cannot fathom a viable model of academic development that does not attend to knowledge in some pivotal way, the possible configurations of cognitive, motivational, and sociocultural

dimensions are extensive and allow for tremendous latitude and creativity for educational researchers (Ackerman, Kyllonen, & Roberts, 1999).

Third, I believe it is important to examine academic development *in situ*, as it unfolds during the interactions between teachers and students, as well as between students and students (Alvermann, Hichman, Moore, Phelps, & Waff, 1998; Cognition and Technology Group at Vanderbilt, 1996). Within this epistemological framework, learning is held to be both individual and social; that is, the *learner* and the *community* are inherently related in formal learning. For this reason, it is essential to consider the social dynamic, even as we consider the variability within and across individuals as they progress from one learning environment or situation to another (American Psychological Association Presidential Task Force on Psychology in Education, 1993; Learner-Centered Principles Revision Work Group, 1995). In effect, studies of group processes should not be cast as antithetical to studies in which individuals are the unit of analysis, but rather as complementary pieces of a very complex puzzle. Said another way, the days of building models of academic development *strictly* from laboratory studies or highly contrived experiments are long past (Alexander, 1996). This is not to disparage the contributions of such approaches, but to call for a model of collaborative, cross-discipline inquiry that values a range of research methodologies and paradigms.

Why Should We Pursue Models of Academic Development?

It is perhaps fair to ask why we should, as a community, aggressively invest in the study of academic development. Do we really need a sequel to *Schooling and the Acquisition of Knowledge*? Personally, I see the study of academic development in terms of a moral imperative. It is quite remarkable to consider that those who read this—despite vast differences in age, gender, ethnicity, or cultural background—share a common characteristic. All have been formally educated. Moreover, this formal educational experience was not brief, but encompassed years, if not decades, of their lives. In addition, even though some have elected to continue their formal education well beyond the period required, many in our society do not

share a similar passion for academic learning and have had to be carried kicking and screaming into the educational experience.

Given that schooling is such a common and requisite part of so many lives, it is startling to realize that we still understand so little about the systematic, long-term outcomes derived from participating in the formal educational process. It is not simply intriguing to study academic development. We really cannot, in good conscience, ignore the ramifications of such a mandatory and pervasive experience in so many lives. For example, do we understand how the specific nature of mathematical instruction received by children in the primary grades not only impacts their conceptions of mathematics, their problem solving abilities, interests in and attitude toward mathematics at that juncture, but also *how* these outcomes impact subsequent development? Further, what are the potential consequences for continued academic development when students succeed at acquiring some body of historical information but manifest no corresponding rise in interest in that domain or improvement in their ability to apply historical thinking in varied contexts? These are just a few of the questions that can be explored under the banner of academic development.

Consider for a moment one startling fact: Despite the tremendous social-political investment that many societies have made in the schooling of their young; despite the rancor and debate over the effectiveness or ineffectiveness of our current educational institutions; despite the decades upon decades of research into learning and human development; despite the plethora of theories dealing with cognitive, physical, social, emotional, or motivational development, we continue to operate as educational practitioners and researchers with virtually no coherent and comprehensive model of academic development. I find this situation unconscionable. Until we begin to articulate models of academic development that encompass hypothesized patterns in learning outcomes, as related to schooling, and that allow for systemic inquiry and evaluation of formal, intentional education, our understanding of the relationship between schooling and knowledge will remain undeniably constrained and regrettably fragmented (Perkins, 1992; Whitehead,

1929/1967). Moreover, judgments about educational effectiveness may become relegated to more superficial and short-term ends (Donmoyer, 1998).

Perhaps one barrier to the aggressive pursuit of coherent and comprehensive models of academic development is the lack of consensus as to the underlying purposes of schooling (Berliner & Biddle, 1995). Clearly, there are multiple purposes for the creation and continued existence of schools (Dewey, 1916/1944). Still, *one* significant purpose of these social/cultural/political institutions we call “schools,” particularly in post-industrial societies, is to transmit critical understandings or knowledge to broad segments of the population (Bereiter & Scardamalia, 1989; Phillips, 1995). This is not to imply that the pursuit of knowledge is the *sole* aim of schools. Still, knowledge remains an expected and valued outcome of formal education and a critical component in any model of academic development.

How Do We Overcome Perceived Barriers?

In addition to the potential lack of consensus on the purpose of schooling, we can probably intone a litany of explanations as to why no models of academic development have yet taken root in this community. Among the possible reasons are

- the unbelievable complexity of learning and development;
- the dynamic nature of human existence; and
- the dramatic range of contextual and sociocultural variability.

Even if we grant such powerful forces, they are the same ones with which we struggle in any of our efforts to research critical educational constructs. Certainly, in the past, when we, as a community, have determined that something matters deeply to us, we have shown astonishing ingenuity and persistence even in the face of formidable obstacles. In addition, there are some initial steps we can take to prepare for such a significant undertaking:

- constitute interdisciplinary teams of researchers and practitioners who are strongly committed to this mission;

- synthesize the wealth of relevant theory and research that already exists, as a way to guide subsequent inquiry;
- employ a range of quantitative and qualitative methodologies that can provide contrasting, but rich and complementary, pictures of academic development; and
- dedicate the substantial financial and human resources to this endeavor that it deserves.

Two additional epistemological/ontological issues are of concern in overcoming barriers to academic development. The first deals with the general pall that has fallen over the construct of formal knowledge within the educational research community, especially in the last decade. As Sfard (1998) suggests, there are those for whom terms such as *knowledge* or *concept* no longer have a place in the educational vernacular. Even less extreme theoretical stances on human learning have come to dwell on the tentativeness, boundedness, or completely individualistic nature of knowledge (Greeno et al., 1998; von Glaserfeld, 1991). Such stances may intentionally or unintentionally call into question the passionate pursuit of knowledge or cast a shadow over efforts to understand the systematic transformation in knowledge resulting from formal education. Why dedicate oneself to seeking that which has so little permanence, generalizability, or certainty?

My second concern relates to this issue as well. If I were to categorize the last decade in the simplest of terms, I would stress its temporal or fleeting quality. This has been a decade seemingly devoted to the "here and now," where ideas can completely absorb us one minute, only to be considered totally irrelevant and forgettable the next. It has also been the decade of process more than product, where *doing* seems to have become an end rather than a means (Sfard, 1998). The human attention span has seemingly grown frighteningly short, even as our thirst for "seductive details" has become almost insatiable (Garner, Gillingham, & White, 1989). It would appear that those theories of learning that emphasize *process* and, knowingly or unknowingly, disparage *product* feed into this moment-to-moment and situation-to-situation mentality. This leaves me to question what students, embarking on the formal educational

journey, should come to expect for the time and effort they invest? What beyond the momentary stimulation, the grades, or the degrees should they expect to show for 12 or more years of education?

Concluding Thoughts

Even in delineating the potential barriers to models of academic development, I must reaffirm my perception that this understanding is worth whatever effort it requires. Even if we believe that our struggle to understand the developmental character of formal education will, at best, result in only a very limited and sketchy portrayal of an immensely complex and dynamic phenomenon, such a condition seems highly preferable to embarking on our educational journeys naively—equipped with only a vague hope that an intellectual promise land lies somewhere over the horizon. It is at times like these that I conjure the images of those intrepid (or foolhardy) voyagers in our ancient past who set out upon the waters of discovery with only crude mappings to guide their way—mappings that occasionally marked the unknown with such daunting phrases as "beyond this point, there be dragons!" I recognize that this call for a concerted effort to chart the course of academic development may likewise be judged as foolhardy by some within this community. But do we have a choice? Can we continue to send our children off on their journeys of formal learning without a far richer and deeper understanding of what they should merit as a result?

I would conjecture that we have *mandated* this educational right-of-passage for our children and youth for some reason and with certain expectations. Perhaps it is because we feel schooling is essential for these learners' ultimate survival, well being, or success in this post-industrial society, or for the very survival of the society itself. Maybe we, like Dewey (1913), believe that the formal educative process will give rise to a body of valued knowledge that will serve us throughout our lifetimes. Whatever our noble purposes, do we not owe it to our children and youth, and to ourselves, to chart this long-term venture and the systematic changes that it *can* or *should* evoke? Is it not time to recommit ourselves to the laudable goals of that Shelter Island conference of 1975 by seeking to under-

stand how schooling relates to knowledge and to knowing? I feel the time for such a commitment is long overdue.

Notes

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ground in educational and cognitive psychology. Send a cover letter, vita, reprints of relevant publications, and two letters of reference to: Professor Robert J. Sternberg, Department of Psychology, Yale University, PO Box 208205, New Haven, CT 06520-8205. Or fax to (203) 432-8317 by March 31, 2000. Yale University is an Equal Opportunity Employer (EOE).

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The California Commission on Teacher Credentialing (CCTC) in Sacramento, CA establishes professional standards, assessments, and examinations for state educator licensing. Two permanent, full-time state employee Consultant positions are available in the Examinations and Research Unit. Consultants perform a variety of professional duties to assure the validity, legal defensibility, and appropriate administration and use of the examinations/assessments relied on by the Commission in licensing decisions. Responsibilities include job analyses, validity studies, and standard-setting studies; monitoring contractors; working with expert advisory panels; and preparing and presenting written reports. \$4,898-\$5,951/month plus benefits. Applicants need (1) an advanced degree in educational research/measurement/evaluation, behavioral sciences, industrial psychology, or closely related field (or CA administration credential); (2) at least three years of professional experience in (a) developing, validating, and managing large-scale testing/assessment programs, especially in education, (b) planning and conducting research studies in education or educational/psychological testing and measurement, and/or (c) teaching educational measurement/research at the postsecondary level; and (3) strong written and oral communication skills. Send resume and two recent examples of professional writing to Bob Carlson, CCTC, 1900 Capitol Ave., Sacramento, CA 95814. Send questions to bcarlson@ctc.ca.gov. CCTC is an equal opportunity employer.

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